POSITIVE DISPLACEMENT FLOW METER

1. Working principle

1.0 Working principle for the metering body

M series meter is positive displacement meter for liquids. They are designed for liquid metering both in transfer and process control applications. Thanks to their design they are easy to keep and can suit a wide range of applications. The meter consists of a housing where two bladed displacement rotors and a central single blocking rotor turn in synchronized relationship within three cylindrical bores with no metal-to-metal contact within the meter element. Each rotor is supported on either end by a bearing plate through which the rotor shafts protrude.

The bladed displacement rotors, alternately move through the two half-cylinder bores of the meter element, while the single blocking rotor rotates within its bore in such a way as to produce a continuous capillary seal between the unmetered upstream product and the metered, downstream product.

At one end of each rotor shaft is a timing gear. The blocking rotor gear, having twice the number of teeth of each of the displacement rotor gears, rotates at half the RPM of the displacement rotors.

2 Model

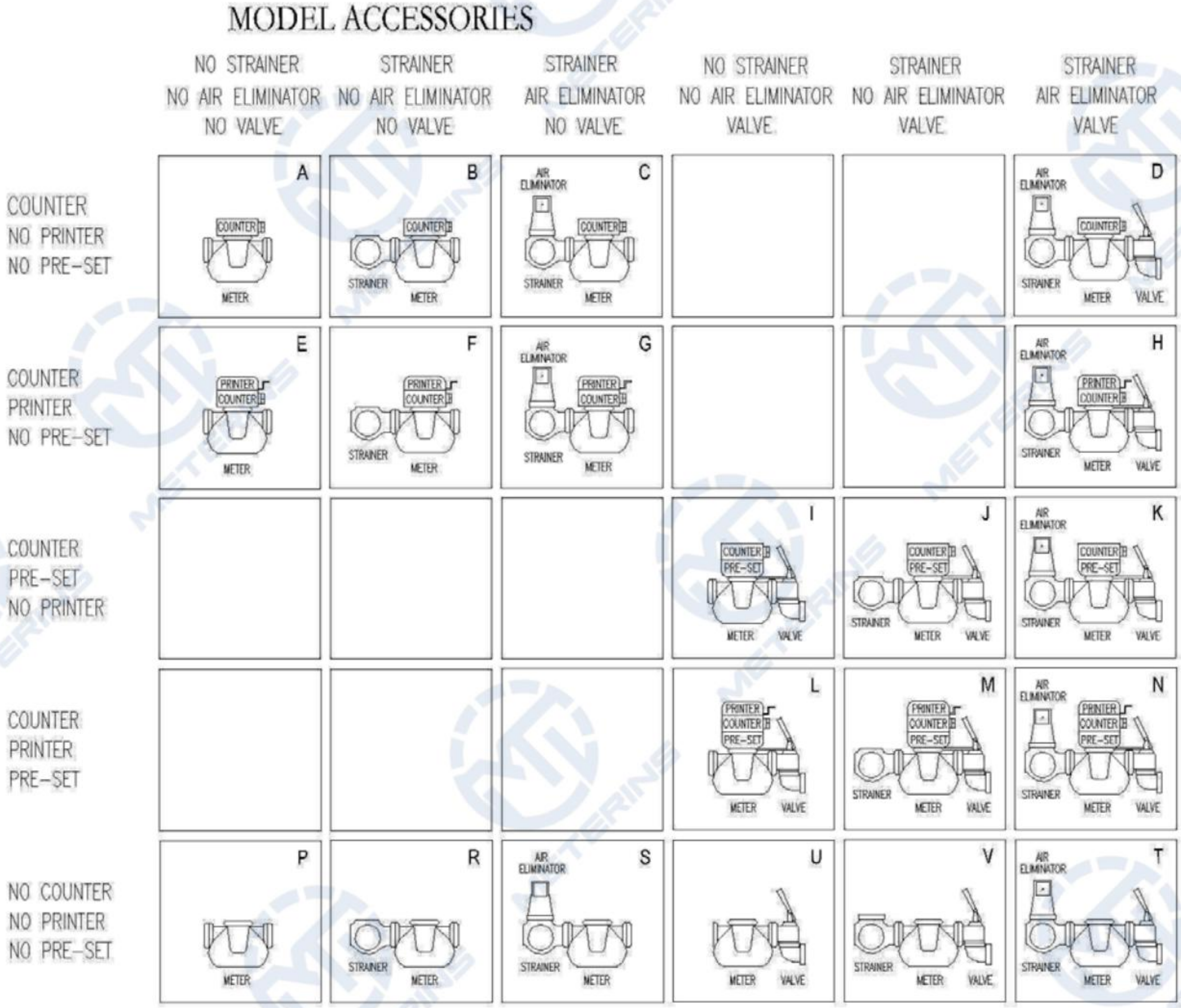
2.1 Model Description

M - 50 - CX - 1

TYPE	NORMAL CAPACITY	MODEL ACCESSORIES	AIR CHECK VALVE	MALERIALS OF CONSTRUCTION
M	50	C	X	1
M METER GLQ STRAINTER V VALVE K CHECK VALVE FL AIE ELIMINATOR	40 60GPM 50 100GPM 80 200GPM 100 350GPM 150 600GPM	A THRUP… SEE CHART OF ACCESSORY COMBINATIONS BELOW	BLANK NO AIR CHECK OR DIFFERENTIAL VALVE X EITHER AIR CHECK OR DIFFERENTIAL VALVE Y EITHER AIR CHECK OR DIFFERENTIAL VALVE PLUS TEMPERATURE VOLUME COMPENSATOR	SEE "METER MATERIALS OF CONSTRUCTION" CHARTS CLASS 1 THRU 20

2.2 Model Accessories

3.0 Typical Application:



- Check of loading/unloading operations of fuel and petrochemical products in fuel bulk plants and/or refineries
- On truck tanker for fuel/LPG transport and delivery
- Heavy duty fuel dispensing system for big vehicles and airplanes

Typical Application of Aluminum Construction

Class 1 meter: refined petroleum products, such as gasoline, fuel oil, diesel fuel, kerosene, ethylene glycol, motor oils and rotogravure ink.

Class 2 meter : aviation gasoline and jet fuels.

Class 3 meter: a wide variety of products such as: liquid sugars, corn syrup, corn sweetners, dextrose, fructose, sucrose, maltose, lactose, corn oil, soy bean oil, cotton seed oil, coconut oil, and shortening's etc. rate of flow is based on viscosity to pressure loss relationship.

Class 10 meter: liquefied petroleum gas(LPG) including butane, isobutene, pentane, ethane, freons and propane.

Class 14 meter: crude oil, also for heated and/or viscous liquids including animals fats, resins, #6 oil and non-abrasive asphalt emulsions.

Class 15 meter: for metering oil or water based latex products, polyester resins, and adhesives(neutral pH). also available for metering herbicides and nitrogen fertilizer solutions (requires Viton and Teflon seals).

Class 16 meter: for general solvent metering, such as methanol, toluene, xylene, naphtha, acetone, MEK, MIBK, and alcohols including ethanol.

4.0 Technical data

Flow rate: M-50-1 2” 50L/min-500L/min
M-80-1 3” 100L/min-1000L/min
Max Pressure: 10Bar
Standard measure unit: Liter and US Gallon for option
Repeatability: 0.05%
Type of flange: ANSI, NPT, BSPF
Strainer mesh: 80Mesh for diesel, 40Mesh for gasoline.